INCH-POUND

MIL-DTL-17361/2A(SH)
19 September 2006

SUPERSEDING
MIL-C-17361/2(SH)
10 March 1989

DETAIL SPECIFICATION SHEET

# CIRCUIT BREAKERS, AIR, ELECTRIC, INSULATED HOUSING (SHIPBOARD USE) TYPES AQB-A100 AND NQB-A100 (FOR REPLACEMENT ONLY)

This document remains inactive for new design.

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17361.

## LIST OF TABLES

TABLE I. Requirements. ....2
TABLE II. Instantaneous trip calibrations (all 3 poles)....3
TABLE III. Tripping characteristics – long-time delay (LTD) (all 3 poles)....4
TABLE IV. Test requirements. ....4

## LIST OF FIGURES

FIGURE 1. Circuit breaker types AQB-A100 and NQB-A100, outline and mounting dimensions. ....5
FIGURE 2. Mounting blocks for circuit breaker types AQB-A100 and NQB-A100 outline and mounting dimensions. ....7



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TABLE I. Requirements.

| | |
|---|---|
| Maximum rated continuous current | 100 amperes |
| Voltage and frequency | 500 VAC, 60 Hz<br>500 VAC, 400 Hz<br>250 VDC |
| Interrupting rating | 15,000 amperes, 60 Hz, AC<br>7,500 amperes, 400 Hz, AC<br>10,000 amperes, 250 VDC<br>7,400 amperes, 355 VDC |
| Peak let-through current | Not applicable |
| Trip unit | Interchangeable |
| Current limiting fuses | Not applicable |
| Type of connections | Front, back (removable mtg.), and combination |
| Electric operator | Not available |
| Undervoltage trip or shunt trip | Neither available |
| Auxiliary switch | Not available |
| Handle lock | Group 2 (Figure 1 of MIL-DTL-17361) |
| Handle boot | Not available |

NOTES:

1. Type AQB-A100 and NQB-A100 circuit breakers shall be used for replacement only.
2. NQB circuit breakers shall consist of an AQB type without trip unit and a modification kit of metallic parts.



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TABLE II. Instantaneous trip calibrations (all 3 poles).

| Trip element rating (amps) /1 | Instantaneous trip settings – amperes/maximum time /2 | | | | | |
|---|---|---|---|---|---|---|
| | 60-Hz AC trip unit | | DC trip unit | | 400-Hz AC trip unit | |
| | Current (amperes) | Time (seconds) | Current (amperes) | Time (seconds) | Current (amperes) | Time (seconds) |
| 15B | 195 | 0.024 | ---- | ---- | ---- | ---- |
| 15BF | 98 | 0.024 | ---- | ---- | ---- | ---- |
| 15BH | ---- | ---- | ---- | ---- | 195 | 0.028 |
| 15BE | ---- | ---- | 98 | 0.028 | ---- | ---- |
| 25B | 325 | 0.024 | ---- | ---- | ---- | ---- |
| 25BF | 163 | 0.024 | ---- | ---- | ---- | ---- |
| 25BH | ---- | ---- | ---- | ---- | 325 | 0.028 |
| 25BE | ---- | ---- | 163.1 | 0.028 | ---- | ---- |
| 50B | 650 | 0.024 | ---- | ---- | ---- | ---- |
| 50BF | 325 | 0.024 | ---- | ---- | ---- | ---- |
| 50BH | ---- | ---- | ---- | ---- | 650 | 0.028 |
| 50BE | ---- | ---- | 325 | 0.028 | ---- | ---- |
| 75B | 975 | 0.024 | ---- | ---- | ---- | ---- |
| 75BF | 488 | 0.024 | ---- | ---- | ---- | ---- |
| 75BH | ---- | ---- | ---- | ---- | 975 | 0.025 |
| 75BE | ---- | ---- | 488 | 0.028 | ---- | ---- |
| 100B | 1300 | 0.024 | ---- | ---- | ---- | ---- |
| 100BF | 650 | 0.024 | ---- | ---- | ---- | ---- |
| 100BH | ---- | ---- | ---- | ---- | 1300 | 0.028 |
| 100BE | ---- | ---- | 650 | 0.028 | ---- | ---- |
| 100BG | 2000 | 0.024 | ---- | ---- | ---- | ---- |

NOTES:

/1 Identification marking of trip units:
Numerals indicate continuous current rating of the specific trip unit.
Letter “B” denotes that the trip units are applicable to circuit breaker type AQB-A100.
Letter “E” denotes calibration for DC applications of type AQB-A100 circuit breaker.
Letter “F” denotes calibration for applications having low inrush currents.
Letter “G” denotes special calibration for use as generator or transformer protection.
Letter “H” denotes calibration for 400-Hz applications.

/2 The tolerance for instantaneous trip shall not exceed ±10%. Time shall be measured at three times pickup current. For qualification testing only, not for routine calibration.



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TABLE III. Tripping characteristics – long-time delay (LTD) (all 3 poles).

| **Current as multiple of trip element rating** | **Tripping time 60-Hz AC trip element DC trip element** | **Tripping time 400-Hz AC trip element** |
|---|---|---|
| 1.35 | ---- | Not less than 1 hour |
| 1.50 | Not less than 1 hour | ---- |
| 2.25 | Less than 1 hour | Less than 1 hour |
| 6.00 | Between 20 and 32 seconds | Between 35 to 45 seconds |

TABLE IV. Test requirements.

<table>
<tr><td>Electrical endurance test</td><td colspan="4">The rate of operation shall be 6 operations per minute for a total of 6000 close-open operations.</td></tr>
<tr><td rowspan="4">Temperature rise tests</td><td colspan="4">Size of cable and bus bars for temperature rise tests</td></tr>
<tr><td rowspan="2">Copper rating (amperes)</td><td colspan="2">Cable size</td><td>Copper bus size (inch)</td></tr>
<tr><td>AC</td><td>DC</td><td>AC and DC</td></tr>
<tr><td>100</td><td>AWG 3</td><td>AWG 3</td><td>3/4 X 1/8</td></tr>
</table>



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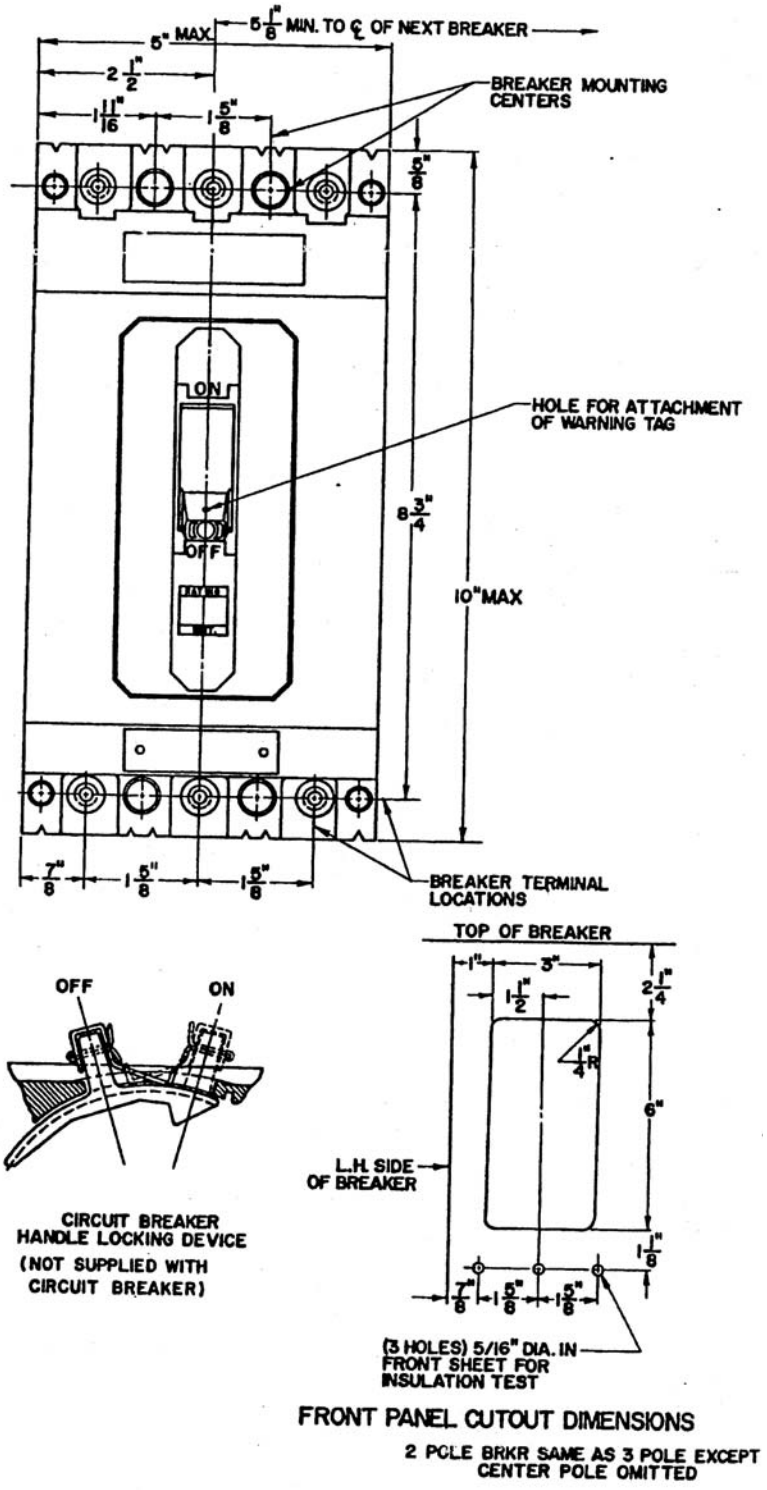



FIGURE 1. Circuit breaker types AQB-A100 and NQB-A100, outline and mounting dimensions.



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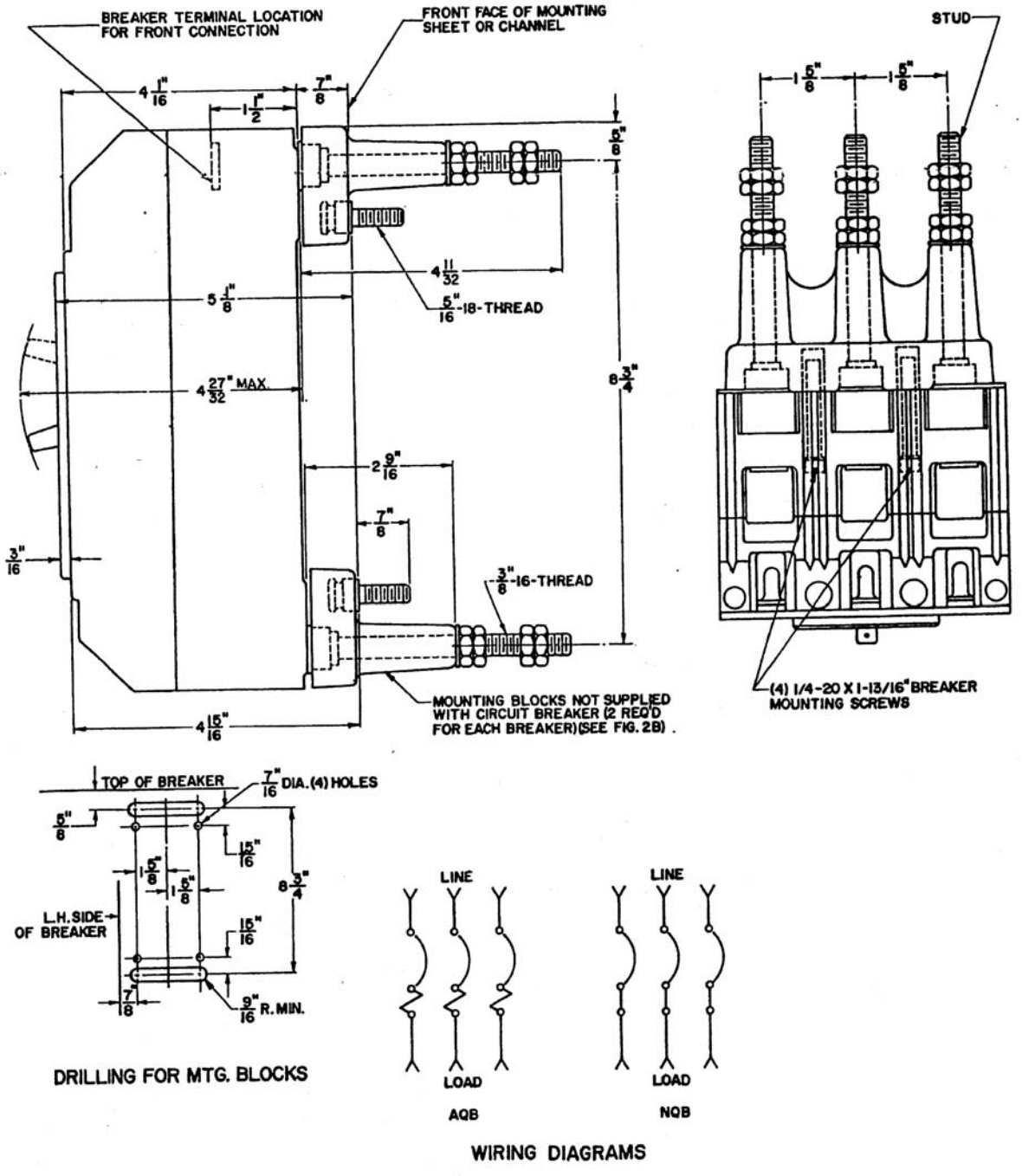
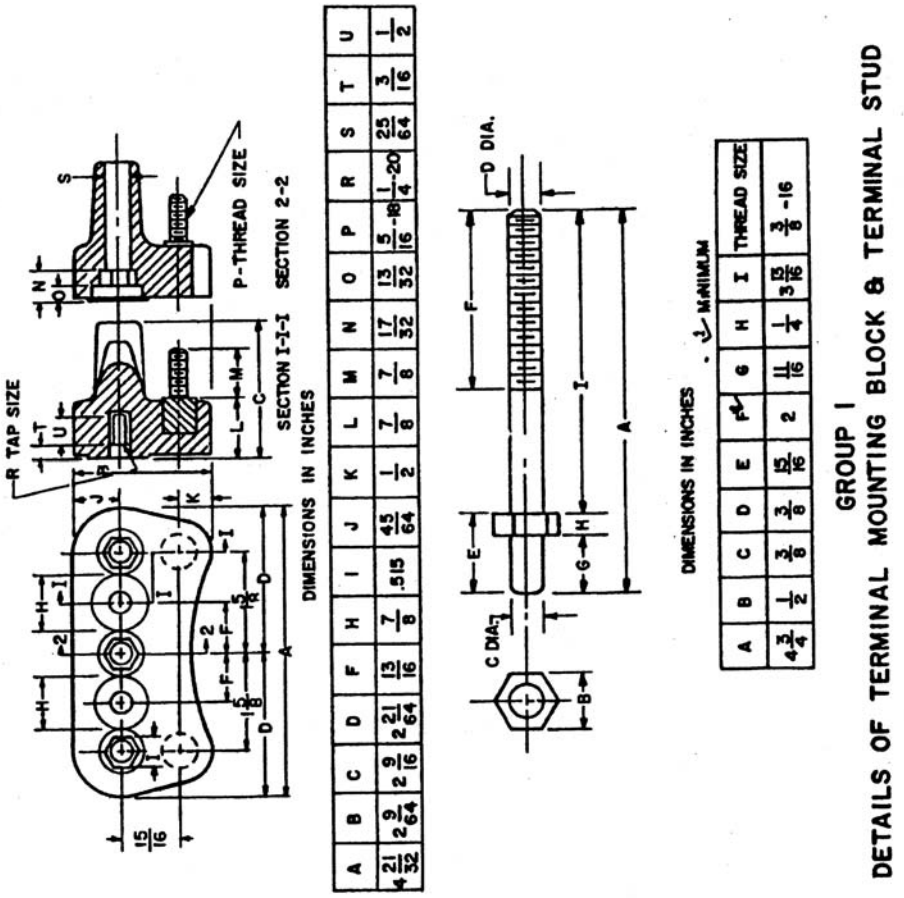



FIGURE 1. Circuit breaker types AQB-A100 and NQB-A100, outline and mounting dimensions - Continued.



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| A | B | C | D | F | H | I | J | K | L | M | N | O | P | R | S | T | U |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 21/32 | 2 9/64 | 2 9/16 | 2 21/64 | 13/16 | 7/8 | .515 | 45/64 | 1/2 | 7/8 | 7/8 | 17/32 | 13/32 | 5/16-18 | 1/4-20 | 25/64 | 3/16 | 1/2 |

| A | B | C | D | E | F | G | H | I | THREAD SIZE |
|---|---|---|---|---|---|---|---|---|---|
| 4 3/4 | 1/2 | 3/8 | 3/8 | 15/16 | 2 | 11/16 | 1/4 | 3 13/16 | 3/8-16 |

FIGURE 2. Mounting blocks for circuit breaker types AQB-A100 and NQB-A100 outline and mounting dimensions.



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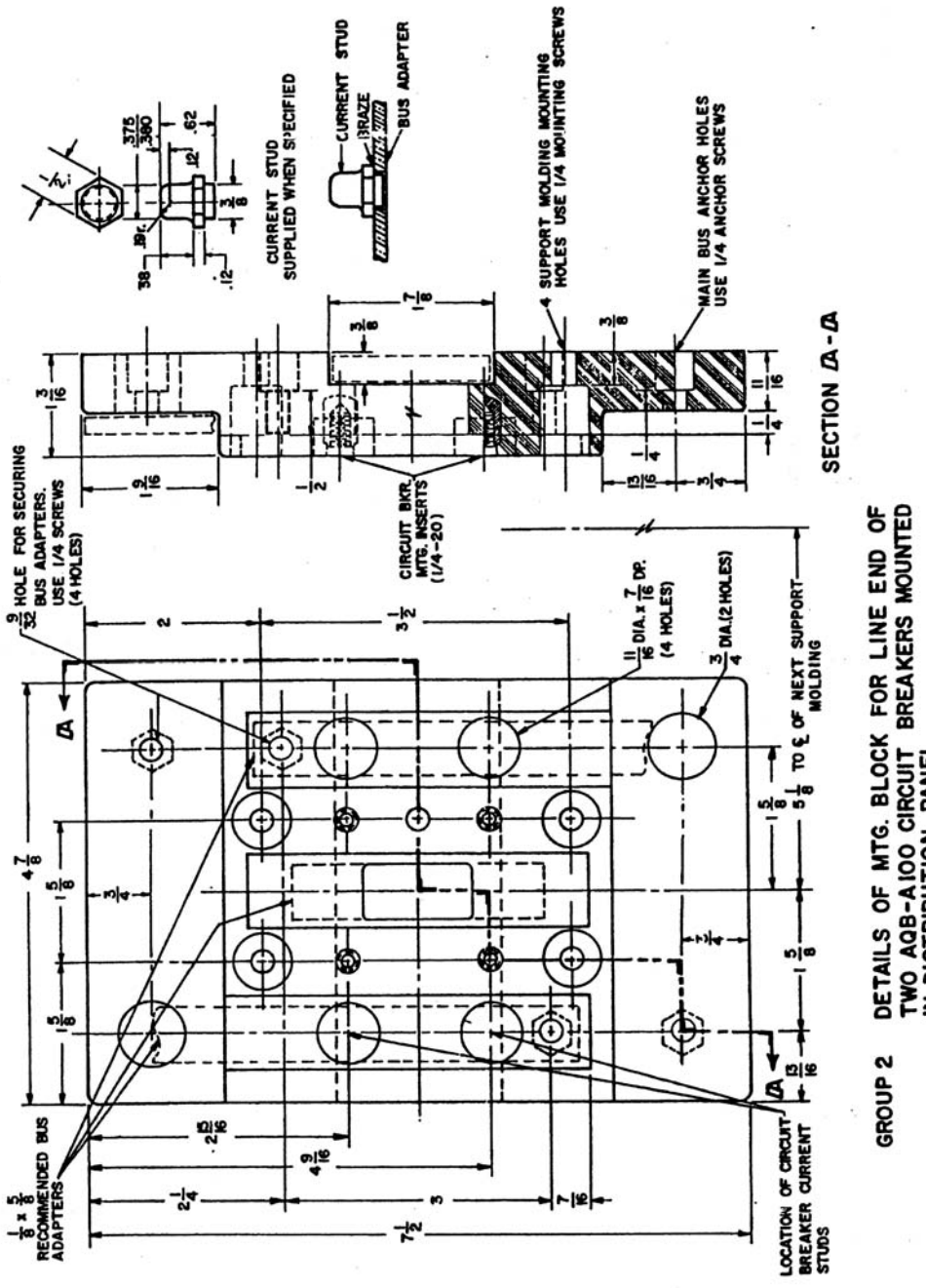



FIGURE 2. Mounting blocks for circuit breaker types AQB-A100 and NQB-A100 outline and mounting dimensions - Continued.



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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing Activity:
Navy – SH
(Project 5925-N185-002)

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